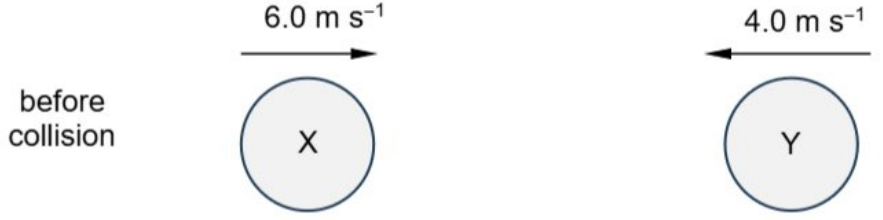
9 Body X of mass 2.5 kg undergoes a head-on collision with body Y of mass 1.0 kg. Bodies X and Y have initial speeds of 6.0 m s^{-1} and 4.0 m s^{-1} respectively.

The two bodies are in contact for a time of 20 ms. After the collision, body X continues in the same direction with speed 2.0 m s^{-1}.



What is the magnitude of the average force exerted on body Y during this collision?

A 10 N **B** 200 N **C** 250 N **D** 500 N